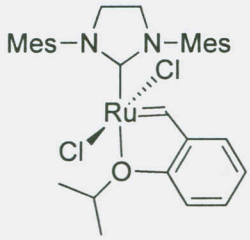
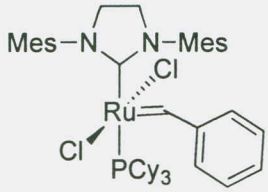
Organic Cume
April 17, 2009

Consider the three olefin metathesis catalysts below.

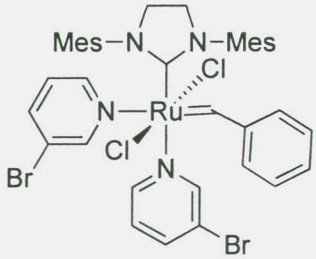
1. Draw the mechanism of the ring-closing-methathesis reaction for the Grubbs II and Grubbs-Hoveyda catalysts, showing each intermediate along the way.
2. After one complete turnover of catalyst, what is different and what is the same?
3. The Grubbs-Hoveyda catalyst gives higher yields than Grubbs II in the cross metathesis between terminal alkenes and acrylonitrile (CH_2=CH–C≡N). Postulate a reason for the difference.
4. How would you expect the G2-pyridine catalyst to behave – more like Grubbs II or more like Grubbs-Hoveyda? Why?



Grubbs-Hoveyda



Grubbs II



G2-Pyridine